
KEY

SCIENCE 24

UNIT A WORKBOOK



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Chapter 1: Chemistry Now and Then

1. Define the following:

a. Synthetic

human-made substance (p. 6)

b. Pesticides

substances that use chemicals to kill plants or animals (p. 6)

c. Herbicides

chemical that kills particular plants (p. 6)

d. Insecticides

chemical that kills particular insects (p. 6)

e. Synthetic Fibers

human-made fibres (such as cloth or thread) made in the laboratory using raw chemicals (p. 8)

f. Synthetic Polymers

human-made materials consisting of long chains of molecules; often spun into thread to make fabrics or molded to form plastics (p. 9)

g. Alloy

two or more metals mixed together; looks like one metal (p. 10)

h. HHPS

Hazardous Household Product Symbols

safety symbols placed on hazardous products to help consumers safely use, store, and dispose of them (p. 12)

i. WHMIS

Workplace Hazardous Materials Information System

provides detailed information on how to store, handle, and dispose of dangerous substances at work (p. 19)

j. MSDS

Material Safety Data Sheet

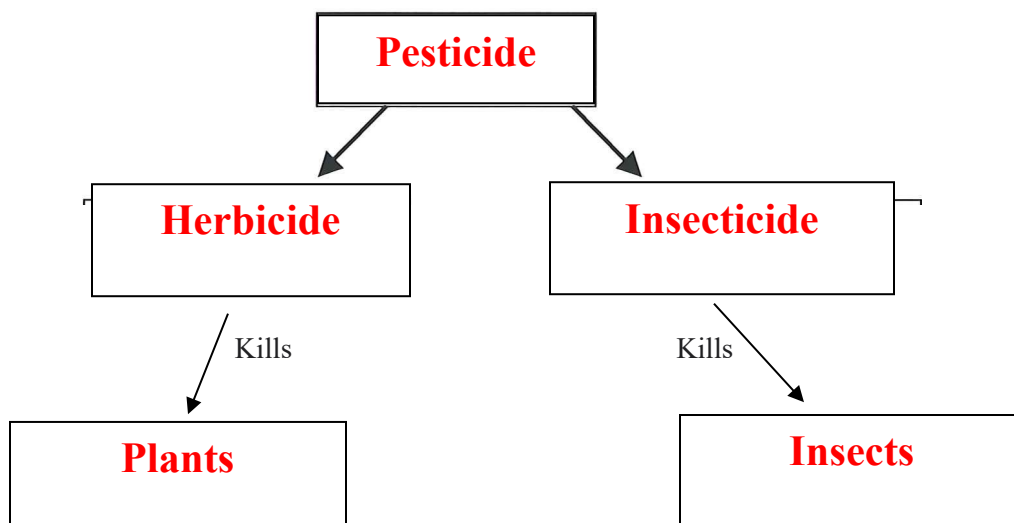
provides detailed information (that will not fit on the product label) about chemicals found in the product (p. 19)

2. Complete the chart using the following 5 words (pg6)

Pesticide
Herbicide

Plants
Insecticide

Insects



3. What does synthetic mean?(pg6)

Human-made substances

4. How do chemists help modern farmers?(pg6)

Synthetic chemicals such as insecticides, herbicides, and fertilizers allow modern farmers to produce a more food with less human labour.

5. How does modern soap differ from soap made 100 yrs ago?(pg7)

- **Instead of using sodium hydroxide, pioneers would run water through the ash pile (ashes from the fireplace) and collect the solution that formed. In fact, they were making their own lye. They would also collect the fat from animals they had slaughtered for food. Nothing was wasted 100 years ago.**
- **Many of today's mass-produced soaps are not really soap at all. They are detergents made from synthetic chemicals. The natural glycerin has been removed and preservatives and other products added to make the bars harder. In commercial soap making, the glycerin is siphoned out, sold, and made into other products.**

6. Name 2 synthetic and 2 natural fibres. Describe at least two advantages of synthetic fibres. (pg 8)

Natural – Cotton, flax, wool , silk

Synthetic - Nylon, rayon, and polyester are three

Benefits include: relatively easy to produce, do not depend on weather or other natural factors to grow, unlimited supply, can be dyed easily, can be treated to meet most human needs.

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7. What are synthetic polymers? (pg 9)

Human-made materials made up of many small, identical sub-molecules.

8. Give two examples of natural polymers and 2 examples of synthetic ones.(pg 9)

Natural – spider web silk, rubber tree sap

Synthetic – neoprene, polyethylene, polystyrene, polyurethane, polycarbonates, nylon, rayon

9. Synthetic polymers are used in many products. Name three synthetic polymers and briefly explain how they have changed daily life.(pg 9)

Examples – see above

Synthetic polymers are developed with physical characteristics that meet a particular need.

Sample polymers and uses are provided in Table 1.1 on page 9

10. In winter you choose clothes with fabrics that will keep you warm. (pg 9)

- a. What materials keep you warm today?

- **Gore-Tex™**
- **polar fleece**
- **fibrefill**

- b. What materials would you have used 100 years ago? (pg 8)

- **wool**
- **down or feathers**

11. What metals are found in stainless steel utensils? How would your life be different if kitchen utensils were not made from stainless steel? (pg 10/11)

Stainless steel is made by combining steel with corrosion-resistant metals such as chromium, nickel, and carbon.

If kitchen utensils were not made of stainless steel, knives would not stay as sharp, forks would bend, and all would rust and have to be replaced.

12. Many household cleaners rely on chemical changes to do their work. What chemical change occurs when you add drain cleaner to a clogged pipe?(pg 12)

Most drain cleaners contain a base (sodium hydroxide) that breaks down proteins (such as skin cells and hair) and animal fats that may be clogging the drain.

13. Yeast helps bread to rise. Explain the chemical change that yeast produces and how this makes bread rise. (pg 13)

Yeast converts sugar into carbon dioxide and alcohol.

The carbon dioxide gas makes baked goods rise.

14. Soap cleans your hair but why won't it make your hair smooth and shiny? (pg 16)

Soap reacts with calcium in water, leaving a scum layer on hair that makes it dull and difficult to comb.

Soap solutions are alkaline (bases), which makes the hair cuticle swell up and get rougher. This leaves hair looking dull. It also appears rough.

Soap may leave behind deposits of minerals (soap scum), which will also make hair dull.

15. Choose two different workplaces and identify five hazardous substances you would find in each one.

Automotive Repair Shop — oil, gas, paints, fillers, cleaners, waxes, polishes

Hair Salon — colorants, curling solutions, shampoos, conditioners, cleaners

Office — inks, toner cartridges, correction fluid, fire retardants in paper, upholstery, and carpets.

Construction Site — demolition explosives, wood preservatives, paints for finishing, and solvents (ammonia) for cleaning up.

16. Why is it important for workers at a restaurant to understand WHMIS?(pg 19)

- **Restaurants use corrosive cleaning solutions.**
- **Workers need to know if materials are flammable and what to do in case of fire.**
- **Food additives are chemicals; chefs have to understand safety issues involved in using these chemicals.**

Chapter 2: Common Reactions in Our Lives

1. Define the following:

a. Chemical Reaction

Transformation that occurs when two or more substances react together, producing one or more new substances with different properties from the original substances (p. 24)

b. Reactants

substances present before a chemical reaction (p. 28)

c. Products

substances present after a chemical reaction (p. 28)

d. Combustion

chemical reaction in which a substance combines with oxygen to produce heat and light; burning (pp. 28, 144)

e. Neutralization

type of reaction during which an acid and a base produce a salt compound and water; the pH of these products is more neutral (closer to 7) than the pH of either reactant (p. 29)

f. Precipitate

solid that forms because of a chemical change (p. 34)

g. Energy Changes

obvious or unnoticeable production or absorption of a form of energy (e.g., sound, light, heat) as part of a chemical change (p. 35)

h. Exothermic Reaction

**any chemical reaction that releases heat (thermal energy) (p. 35)
Feels warm/hot.**

i. Endothermic Reaction

**any chemical reaction that absorbs heat (thermal energy) (p. 35)
Feels cool/cold.**

2. List the steps that happen when a banana ripens.(pg 24)

bananas release natural ripening gas known as ethylene

ethylene triggers enzyme production

enzymes speed up chemical reactions

enzyme amylase breaks down starches into sugar

other enzymes trigger colour change and softening of fruit

enzymes break down large molecules into smaller ones that evaporate in the air, providing the banana aroma

3. What is responsible for the color change when using bleach? (pg 24)

Oxygen is responsible for the color change when using bleach

4. Sunlight can break down sodium hypochlorite (bleach). Observe the plastic containers used for bleach. Why will you never see a transparent bleach container?

exposure to light degrades bleach, producing poisonous chlorine gas.

5. How is baking bread a chemical reaction? (pg 25)

Ingredients react chemically to produce a loaf of fluffy bread.

When flour is mixed with water and yeast, enzymes in the yeast break down the starch into glucose.

Other enzymes react with some of the glucose molecules to produce carbon dioxide and ethanol.

6. What safety equipment should you use when handling chlorine in a pool?(pg 25)

Protective rubber gloves and eye protection should be worn when working with chlorine.

7. You have a glowing splint and it reignites in a test tube. What gas is in the test tube?

Oxygen

8. Label the reactants and products, and state whether they are a neutralization or a combustion reaction in the following equations. (pg 29)

a. Magnesium hydroxide and Hydrochloric acid combine to form water and magnesium chloride. **neutralization**

Reactants - Magnesium hydroxide and Hydrochloric acid

Products – water and magnesium chloride

b. Methane and oxygen combine to form carbon dioxide and water.

Combustion

Reactants – methane and oxygen

Products - carbon dioxide and water

9. Give three examples of combustion reactions that you encounter in a day. List as many products and reactants that you can. (pg 29)

Combustion Reaction (reactant)	Reactant	Product(s)
barbecues use propane (C₃H_{8(l)})	oxygen gas (O_{2(g)})	carbon dioxide (CO_{2(g)}) and water vapour (H₂O_(g))
lighters use butane (C₄H_{10(g)})	oxygen gas (O_{2(g)})	carbon dioxide (CO_{2(g)}) and water vapour (H₂O_(g))
vehicles use octane (C₈H_{18(l)})	oxygen gas (O_{2(g)})	carbon dioxide (CO_{2(g)}) and water vapour (H₂O_(g))
racing vehicles use ethanol (C₂H₅OH_(l))	oxygen gas (O_{2(g)})	carbon dioxide (CO_{2(g)}) and water vapour (H₂O_(g))
home heating uses oil or methane (CH_{4(l)})	oxygen gas (O_{2(g)})	carbon dioxide (CO_{2(g)}) and water vapour (H₂O_(g))
candles burn (C₂₅H_{52(s)})	oxygen gas (O_{2(g)})	carbon dioxide (CO_{2(g)}) and water vapour (H₂O_(g))
wood burns (C₁₂H₂₂O_{11(s)})	oxygen gas (O_{2(g)})	carbon dioxide (CO_{2(g)}) and water vapour (H₂O_(g))

10. Fishy odours on your hands are caused by a base and can be eliminated with lemon juice.

Is this an example of a combustion or a neutralization reaction?

Why? What are the reactants? (Pg 29)

This is a neutralization reaction as the odour of the fish is lessened by the addition of an acid.

The reactants are the lemon juice (containing citric acid) and the odour-causing agent of the fish.

11. List 5 things you could observe to tell if there was a chemical change.(pg 34)
- **bubbles/gas forms**
 - **colour change**
 - **energy is released/absorbed**
 - **odour change**
 - **precipitate forms**
12. In the following scenarios state whether a chemical change occurred and state why you know it did. Use the reasons from the previous question
- a. Hydrochloric acid is added to Magnesium and Hydrogen gas bubbles out.
Yes, bubbles/gas formed
- b. Two liquids are mixed together and a solid forms at the bottom of the test tube
Yes, precipitate formed
- c. Phenolphthalein, a colorless indicator, changes color in the presence of an acid
Yes, color change
- d. Water freezes and forms ice
No, this is a phase change, $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{s})$
- e. Iron is heated until it becomes molten
No, this is a phase change, $\text{Fe}(\text{s}) \rightarrow \text{Fe}(\text{l})$
- f. Two chemicals are mixed together, and you can smell spearmint
Yes, odor produced
- g. Two liquids are mixed together and ice forms on the outside of the test tube
Yes, energy is absorbed

13. How do exothermic and endothermic reactions differ? How are they similar? (pg 35)

Both endothermic and exothermic reactions involve energy as evidence of a chemical reaction.

Endothermic reactions involve adding energy to the reactants before the reaction will proceed.

Exothermic reactions involve the release of energy after the reaction has occurred.

14. Some exothermic reactions can be dangerous. Give the reason behind this statement. (pg 35)

- **Exothermic reactions may involve very large amounts of energy. This energy may take several forms, including heat.**
- **The combustion of various materials could lead to forest or house fires.**
- **Explosives rely on massive amounts of energy being released very quickly. If proper safety precautions are not in place, serious injury or even death may result.**

15. For the following situations determine if the reaction is exothermic or endothermic.

- Methane is burnt. **Exothermic**
- When mixing two gases a loud explosion is heard. **Exothermic**
- Sodium acetate reacts and creates ice **Endothermic**
- A plant absorbs energy from the sun **Endothermic**
- Glucose in your cell produces energy. **Exothermic**

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Chapter 3: Types of Chemical Reactions

1. Define the following:

a. Word Equation

symbolic representation of a chemical reaction that uses words to indicate reactants and products (p. 42)

b. Simple Composition

reaction in which two or more elements combine to form a compound (p. 44)

c. Simple Decomposition

reaction that breaks a compound into the elements that make it up (p. 45)

d. Chemical Name

standard scientific name by which each element or compound is known; established by an internationally recognized naming system (p. 50)

e. Chemical Formula

standard scientific symbol by which each chemical element or compound is known (p. 50)

f. Common Name

name by which a common chemical is referred to in society; varies from region to region; existed long before an international naming system was created (p. 51)

g. Law of Conservation of Mass

matter can be neither created nor destroyed (p. 52)

h. Balanced Chemical Equation

reaction equation that has the same number of atoms of each element on both sides of the equation (p. 53)

i. Coefficients

numbers placed in front of chemical formulas to balance the number of atoms on either side of the equal sign in a chemical equation (p. 54)

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2. For each word equation write what you would say when reading it. (pg 42/43)

a. Solid zinc + hydrochloric acid $\rightarrow$ hydrogen gas + zinc chloride

solid zinc and hydrochloric acid produce hydrogen gas and zinc chloride

b. Hydrochloric acid + sodium acetate $\rightarrow$ acetic acid + sodium chloride

hydrochloric acid and sodium acetate produce acetic acid and sodium chloride

3. For each of the following reactions identify whether it is a simple composition reaction, a simple decomposition reaction, a combustion reaction or a neutralization reaction. (pg 44 to 49)

a. Methane + oxygen $\rightarrow$ carbon dioxide + water + energy **Combustion**

b. Hydrogen gas + oxygen gas $\rightarrow$ water **Simple Composition**

c. Water $\rightarrow$ hydrogen gas + oxygen gas **Simple Decomposition**

d. Hydrochloric acid + sodium hydroxide $\rightarrow$ salt and water **Neutralization**

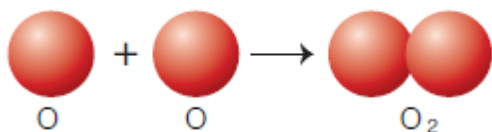
e. $\text{NaCl} \rightarrow \text{Na} + \text{Cl}$ **Simple Decomposition**

f. $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$ **Combustion**

g. Solid sodium hydroxide + liquid acetic acid $\rightarrow$ sodium acetate (a salt) + water **Neutralization**

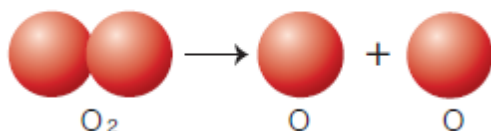
4. Draw a diagram representing a composition reaction. (pg 45)

simple composition reaction



5. Draw a diagram representing a decomposition reaction. (pg 45)

simple decomposition reaction



6. Why do scientists use chemical formulas? (pg 50)

Because scientists speak many different languages, they created a common language that would allow them to communicate with one another.

7. If something is dissolved in water what subscript should you use? (pg 50)

(aq) – means aqueous (dissolved in water)

8. What is the common name for the following compounds?(pg 51)

a. NaCl **table salt**

b. $\text{CaCO}_{3(s)}$ **chalk, limestone, marble**

c. $\text{C}_2\text{H}_5\text{OH}_{(l)}$ **alcohol**

d. $\text{C}_6\text{H}_{12}\text{O}_6(s)$ **sugar**

e. $\text{H}_2\text{O}_{(l)}$ **water**

f. $\text{NaHCO}_{3(s)}$ **baking soda**

9. If I have 10g of reactants how many grams of product should I have? Why? (pg 52)

10g – matter is not created or destroyed in a chemical reaction

10. The following reaction takes place: (pg 52)



a. What would the mass of the reactants be if there was 50g of product?

50g

b. If there was 40g of HCl how much NaOH was there?

50g – 40g = 10g

11. Balance the following equations by filling in the correct coefficient:



Na - 2

Cl - 2

Na - 2

Cl - 2

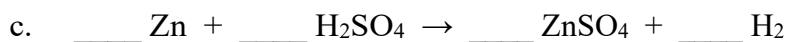


Fe - 4

O - 3 x 2 = 6

Fe - 2 x 2 = 4

O - 2 x 3 = 6



Zn - 1

H - 2

S - 1

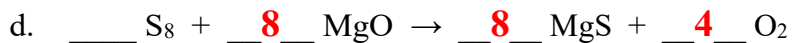
O - 4

Zn - 1

H - 2

S - 1

O - 4



S - 8

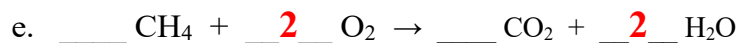
Mg - 8

O - 8

S - 8

Mg - 8

O - 4 x 2 = 8



C - 1

H - 4

O - 2 x 2 = 4

C - 1

H - 2 x 2 = 4

O - 2 + 2 = 4

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Chapter 4: Reactions and the Environment

1. Define the following:

a. Greenhouse Gases

Compounds (such as carbon dioxide, water, and methane) that trap heat in Earth's atmosphere (p. 62)

b. Global Warming

**long-term trend toward warmer temperatures accompanied by extreme climatic events and shifting of the climatic zones;
more correctly called global climate change; see global climate change (p. 62)

formerly called global warming; during this process, some places on Earth may get much warmer while others may get cooler; see climate change (p. 149)**

c. Climate Change

**natural process of Earth's warming and cooling intervals;
see also global climate change (p. 62)**

d. Pollutants

chemicals in the air, water, or soil that cause harm to living things or to the environment (p. 64)

e. Acid Precipitation

Sometimes referred to as acid rain; rain, snow, fog, or dew that has become more acidic; see acid deposition (p. 64)

f. Acid Deposition

acidic materials (liquids or solids that contain sulfuric or nitric acid) that are deposited on Earth (pp. 64, 148)

g. Sulfur Dioxide $\text{SO}_2(\text{g})$

acidic gas that mixes with water vapour in clouds to form acid precipitation (p. 64)

h. Nitrous Oxide $\text{NO}_x(\text{g})$

naturally produced acidic gases that mix with water vapour in clouds to form acid precipitation (p. 64)

i. Liming

**process of adding natural limestone to lake water to neutralize lakes that have been acidified by acid deposition;
neutralizing means bringing the pH of lake water closer to normal (p. 66)**

j. Scrubber

device installed in industrial smokestacks to remove sulfur and other pollutants formed during the combustion of coal, for example (p. 68)

k. Corrosion

process that chemically breaks down or degrades metal (p. 69)

l. Rusting

one type of corrosion (p. 69)

m. Galvanization

process of applying a zinc coating to iron or steel to provide a protective barrier (against rust) between the metal and the environment (p. 72)

n. Sacrificial Metal

**a metal used to protect a nearby metal that is more reactive;
it “draws” the corrosion; magnesium is a common sacrificial metal (p. 73)**

2. What is the reaction we use to convert fossil fuels into energy? Pg 60

Combustion

3. What are the products of combustion? pg 60

Carbon dioxide gas ($\text{CO}_{2(g)}$) + Water Vapor ($\text{H}_2\text{O}_{(g)}$) + energy (heat)

4. How does combustion differ from Incomplete combustion? Pg 60 and 61

Incomplete combustion occurs when there is not enough oxygen available

Combustion produces three products

Incomplete combustion produces five products

5. Name 4 products of incomplete combustion. Pg 61

1. carbon monoxide gas – $\text{CO}_{(g)}$

2. carbon (black soot) – $\text{C}_{(s)}$

3. carbon dioxide gas – $\text{CO}_{2(g)}$

4. water vapor - $\text{H}_2\text{O}_{(g)}$

5. energy (heat)

6. What causes incomplete combustion to occur? Pg 61

Incomplete combustion occurs when there is not enough oxygen available

7. Why can incomplete combustion be deadly? Pg 61

Carbon monoxide gas ($\text{CO}_{(g)}$) is an odourless, colourless gas that is extremely poisonous.

It combines with blood in the bloodstream to starve the body of oxygen, causing death.

8. List 3 natural greenhouse gases. Pg 62

carbon dioxide gas – $\text{CO}_{2(g)}$

water vapour – $\text{H}_2\text{O}_{(g)}$

methane gas – $\text{CH}_{4(g)}$

9. Describe three negative effects of global climate change Pg 62

- **Glaciers are melting.**
- **Climate zones are shifting**
- **Could cause a change in precipitation patterns.**
- **Sea level are rising.**
- **Plant and animal species may not be able to adapt to the climate changes and may become extinct.**

10. What are the major contributors to acid deposition? Pg 64

Human-made pollutants – from vehicles and industry

Natural gas emissions – from volcanoes and forest fires

11. Name two gas emissions that are linked to acid deposition. Pg 65

sulfur dioxide - $\text{SO}_{2(g)}$

nitrogen oxides – $\text{NO}_{x(g)}$

nitrogen monoxide – $\text{NO}_{(g)}$

nitrogen dioxide - $\text{NO}_{2(g)}$

12. What is the acid in the stomach that is neutralized? Pg 66

Hydrochloric acid – HCl(aq)

13. What chemical neutralizes the acid in your stomach? Pg 66

Mucous contains a base that neutralizes any acid that comes into contact with the wall of the stomach.

pancreas releases sodium bicarbonate solution into the small intestine

14. What are the products of a neutralization reaction? Pg 66

A salt and water

15. What is liming and why do we do it? Pg 66

Natural limestone ($\text{CaCO}_{3(s)}$) is a base used to neutralize lakes that have been acidified by acid precipitation.

16. What is the pH range of an acid? Pg 67

0-6.9

17. How can we reduce the amount of acid deposition? Pg 67

A better way to reduce acid deposition is to prevent it from forming.

One way to reduce sulfur emissions is to remove the sulfur from the fuel before the fuel is burned.

Scrubbing is a chemical and physical process that removes sulfur compounds formed during the combustion of coal.

Scrubbers remove the sulfur from the smokestack before it reaches the atmosphere.

18. What are scrubbers? What do they do? Pg 68

Scrubbing is a chemical and physical process that removes sulfur compounds formed during the combustion of coal.

Scrubbers remove the sulfur from the smokestack before it reaches the atmosphere.

19. Describe how driving a car contributes to the problem of acid precipitation. Pg 65

Driving a car requires the combustion of fossil fuels.

This process releases pollutants that contribute to acid deposition.

20. Why is burning coal causing acid deposition? Pg 68

Burning coal releases sulfur dioxide and nitrogen oxides, which react with rainwater to make acid deposition.

21. Why is the corrosion of iron and steel a serious problem? Pg 69

Corrosion of iron and steel is a serious problem because these metals are widely used.

As rust flakes off an iron structure, more metal is exposed to the environment.

The structure is weakened over time, as more and more metal is exposed.

Unless rusting is prevented, it will weaken an iron or steel structure.

The rusting of the steel used to reinforce concrete in bridges and buildings can cause these structures to fail and collapse.

Repairing or replacing these structures costs millions of dollars.

22. What type of reaction is rusting? Pg 69

Simple composition

23. List some properties of iron (III) oxide (Rust). Pg 69

Rust flakes as it forms.

As rust flakes off an iron structure, more metal is exposed to the environment and more rusting occurs.

24. Why does metal rust even after its outer layers have corroded? Pg 69

Rust flakes as it forms.

As rust flakes off an iron structure, more metal on the inner layers is exposed to the environment and more rusting occurs.

25. List 3 ways we can prevent rusting. Pg 72 and 73

Painting - gives a protective coating to prevent oxygen and water from reaching the metals surface

Galvanizing - process of applying a zinc coating to iron or steel that provides a protective barrier between the metal and the environment.

Sacrificial Metal - attaching a sacrificial metal such as magnesium to the iron object, so the sacrificial metal will corrode instead of the iron

26. Why does galvanization work? Pg 72

- the zinc acts like paint, isolating the steel from the environmental factors, like oxygen and moisture, that could cause it to rust; and**
- the zinc acts as a sacrificial metal if the nail is scratched**

Meet with your teacher to go over this unit and pick up a Practice Test

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